

The background is a complex watercolor composition. It features large, soft-edged washes of color in shades of orange, red, pink, and grey. Interspersed among these are numerous small, circular and oval shapes, some of which are dark and textured, resembling planets or celestial bodies. There are also areas of fine, golden-yellow speckling, particularly in the upper left and lower right. The overall effect is artistic and textured, with a hand-painted feel.

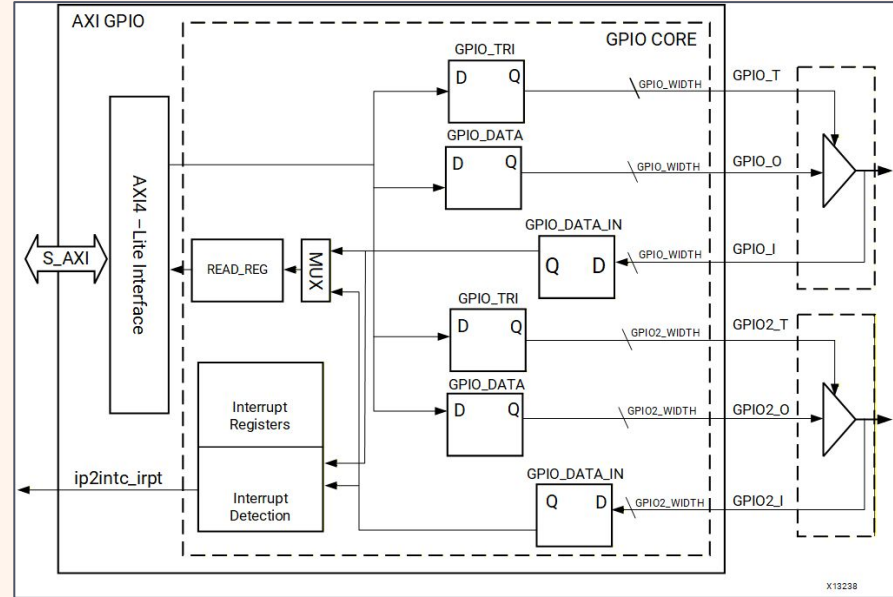
AXI GPIO

By Anna Bagdishyan

ECE 420, Instructor Philip Tracton

Overview

- I chose this block because it was a good way to learn AXI4-Lite while working with a GPIO concept I already understood.
- AXI GPIO is a simple input/output IP block that can be controlled through AXI4-Lite.
- Common applications include driving LEDs and monitoring switches or buttons.



Basic Operation & Key Features

- The AXI GPIO block connects GPIO pins to an AXI4-Lite interface.
- The GPIO_DATA register is used to read input values and write output values.
- The GPIO_TRI register controls the direction of each pin.
 - 0 = output, 1 = input
- The block can be configured as single-channel or dual-channel.
- The GPIO width can be selected from 1 to 32 bits.
- Optional interrupt support can be enabled.

Design, Verification, and Challenges

```
-- Test 5: reading input value 1100
gpio_io_i_tb <= "1100";
wait until rising_edge(clk_out_tb);
wait until rising_edge(clk_out_tb);
wait until rising_edge(clk_out_tb);
wait for 1 ns;
```

```
s_axi_araddr_tb <= GPIO_DATA_ADDR;
s_axi_arvalid_tb <= '1';
s_axi_rready_tb <= '0';
```

```
wait until rising_edge(clk_out_tb) and s_axi_rready_tb = '1';
s_axi_arvalid_tb <= '0';
```

```
-- Finish AXI read
```

```
s_axi_rready_tb <= '1';
wait until rising_edge(clk_out_tb) and s_axi_rvalid_tb = '1';
wait for 1 ns;
```

```
if s_axi_rdata_tb(3 downto 0) /= "1100" then
    report "FAIL: readback should be 1100";
    finish;
end if;
```

```
-- Test 2: writing 1010
```

```
s_axi_awaddr_tb <= GPIO_DATA_ADDR;
s_axi_wdata_tb <= x"0000000A";
s_axi_wstrb_tb <= "1111";
s_axi_awvalid_tb <= '1';
s_axi_wvalid_tb <= '1';
s_axi_bready_tb <= '0';
```

```
wait until rising_edge(clk_out_tb) and s_axi_awready_tb = '1' and s_axi_wready_tb = '1';
s_axi_awvalid_tb <= '0';
s_axi_wvalid_tb <= '0';
```

```
-- Finish the AXI write
```

```
s_axi_bready_tb <= '1';
wait until rising_edge(clk_out_tb) and s_axi_bvalid_tb = '1';
s_axi_bready_tb <= '0';
```

```
wait until rising_edge(clk_out_tb);
wait for 1 ns;
```

```
if gpio_io_o_tb /= "1010" then
    report "FAIL: gpio_io_o should be 1010";
    finish;
end if;
```

